

Assignment 2 COEN 6311 Summer 2024

Group Size of Two or Three

This assignment practices the data model, code review and continuous development within the context of DepartmentDirect. The context continues the assignment 1's scenario "the scenarios of subscribe to future updates and publish event notifications." In the task 5 of assignment 1, two subtasks are selected for prototype. "for the selected user story, break it down into tasks and assign the tasks into the project management system. Select two tasks for implementation and integrate the task development with code IDE, repository and project management system. Please demonstrate the unit testing is passed before the commitment to code repository and project management system reflects the change of states. The states should show the time tracking of the development of the two tasks." Now the requirements further evolve with system architecture design and development requirements.

Requirement (RQ1) : Consider the MVC architecture in which M represents the model, which includes the data model, storage and queries of data represented by the data model.

RQ1.1: Define the data model following JSON structure of two data types, namely Events and Subscribers.

RQ1.2: Apply NoSQL database such as MongoDB to create two collections, namely events and subscribers. The events and subscribers are linked by event IDs and a list of subscribers on a certain event.

RQ1.3: Generate 20 samples for each data type and insert to the data collection.

Requirement (RQ2): Consider the MVC architecture in which C represents the control of business logic. The observer design pattern is an option for structuring the relation and interactions for components involved in subscription and publication. Given the development of the observer pattern practices at the lecture time, now the development should be further integrated with data models involving the two data types.

RQ2.1: Select the tool-based review method and follow pair programming within the assignment team go through the first iteration (1) make the necessary changes, perform unit testing and commit to the repository and create the first pull request; (2) contact project members who are outside the assignment team to pull request and add comments after the review; (3) make the second round changes and unit testing. Create the 2nd pull request for the reviewers' approval; (4) contact the project member for approval in the repo.

RQ2.2 : It is not necessary to develop the view in the MVC architecture for this assignment. Unit testing should verify the following functions.

1) a subscriber subscribes to a certain event with the event ID. The subscriber's ID should be added into event's subscriber list. Such information is updated in the data storage in the MongoDB.

2) given an event, publish the event to all the subscribers. Each subscriber should print out the summary of the event.

Continue the report in assignment one with new sections to address changes in requirements following code review process.

Section C4 Models. Consider C4 models and select the proper level of models to address RQ1 and RQ2.

Section Development. Then develop the necessary program code and unit tests.

Section Code Review. Please document the code review activities to address RQ1 and RQ2. Please write the name of the reviewer and the contribution of the reviewer (reviewer with contributions to the code review will receive extra bonus points).

Please use necessary screenshots for code changes and state changes in the review tools. Please put the github link in the report. The writing should follow the template report is from <https://www.ieee.org/conferences/publishing/templates.html> or IEEE Conference Template - Overleaf, Online LaTeX Editor. Mis-formatted documents will result in not being marked.